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Decentralized Document Verification on Bitcoin Using Taproot

Master Thesis Proposal

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Introduction

Verification of a Document

Organizations issue documents to certify that a specific fact has occurred. These documents may be physical or digital.

In current practice, document verification is usually performed by the issuing organization itself.

This verification requires checking three fundamental properties:

- The content has not been altered.
- The document was issued by the claimed organization.
- The document has not been revoked.

Example: Academic Certificate

An academic certificate illustrates the requirements of document verification:

- Integrity: The content (names, dates, degree) must not be modified.
- Authenticity: The certificate must be issued by the legitimate institution.
- Validity: The certificate may become invalid if revoked due to error or fraud.

Introduction

Verification Problem

Remark

"The global market for certificate forgery and fraud is estimated to exceed two trillion USD annually." [1]

Document forgery directly undermines trust in issuing institutions.

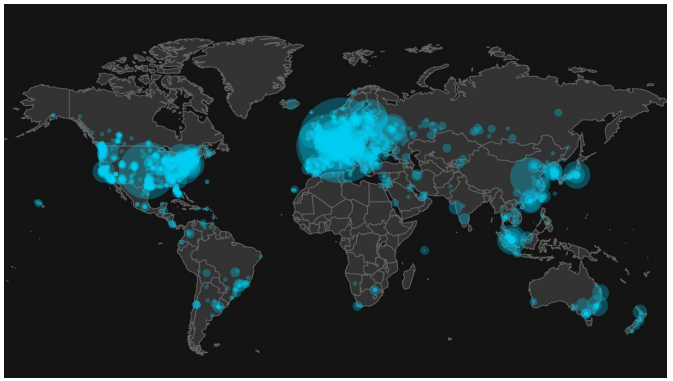
In traditional physical systems, document verification is centralized and requires significant operational resources, including personnel, infrastructure, and anti-tampering measures.

In digital systems, verification remains centralized and introduces additional costs related to service availability, Denial-of-Service protection and server-side cybersecurity.

Introduction

Decentralized Networks

Bitcoin is the first decentralized network, operating continuously since January 3, 2009.
It is secured by a **Proof of Work (PoW)** consensus protocol and elliptic-curve cryptography.



Its main properties are:

- Censorship resistance and permissionless participation
- High availability through global replication.
- An immutable public ledger.
- Public verifiability without trusted intermediaries.

Introduction

Why Bitcoin?

Bitcoin is designed to minimize the attack surface of the network. Reduces the risk of Denial-of-Service (DoS) attacks by limiting on-chain computation.

This is achieved through:

- Stack programming language.
- A small and fixed set of opcodes (e.g., arithmetic, comparison, signature checks).
- Does not allow recursion, loops, or persistent memory.

Using the Unspent Transaction Output UTXO model, each transaction is independent.

Introduction

Blockchain Networks with complex smart contracts

Other blockchain platforms, such as Ethereum, adopt an account-based model instead of UTXO.

They execute programs on the Ethereum Virtual Machine (EVM), enabling expressive logic with loops, recursion, and access memory.

The Trade-off

Increased expressiveness comes at the cost of higher execution fees, greater system complexity, and a larger attack surface compared to Bitcoin's constrained execution model.

Introduction

Decentralized Networks for Document Verification

Decentralized blockchain networks enable new models for digital document verification.
Instead of relying on a single organization to validate documents, verification can be performed independently using public blockchain data.

For document verification, this model provides:

- High availability without dependence on centralized infrastructure.
- Resistance to denial-of-service and censorship attacks.
- Immutable records that preserve data integrity over time.
- Cryptographic proofs that prevent repudiation by the issuer

Literature Review

Beyond Monetary Transactions

Blockchain has been studied as a tool for preserving non-monetary information by Lemieux [2].

Shows that public blockchains can provide integrity guarantees for the preservation of digital records.

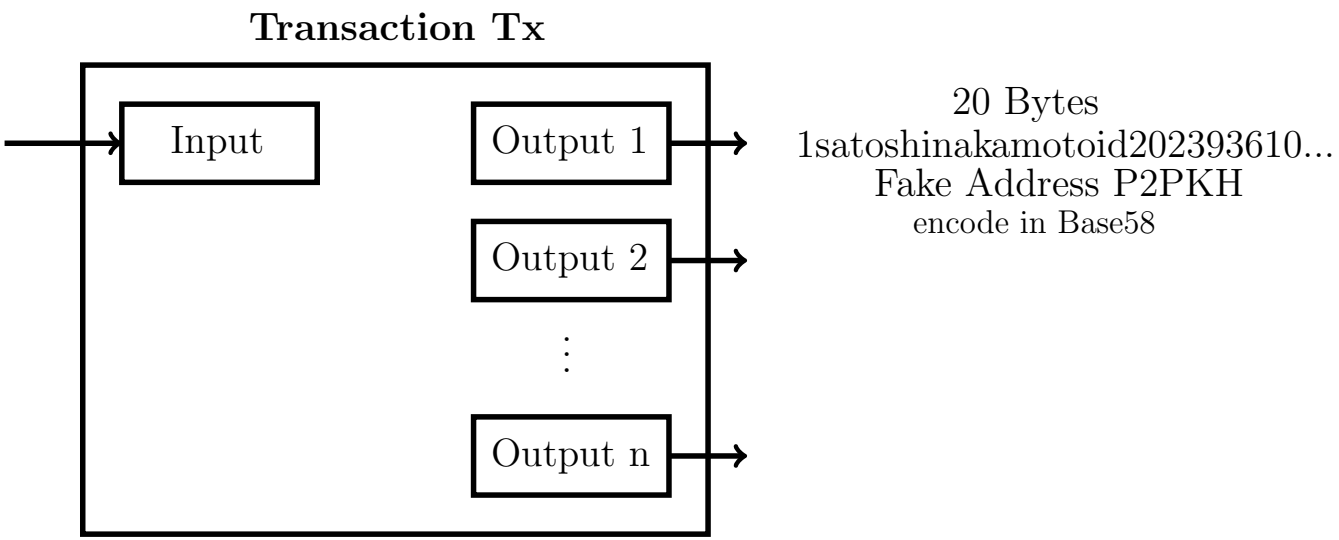
Remark

Blockchain preserves record integrity, but does not guarantee record correctness or long-term archival reliability, since its security relies on hash functions and digital signatures whose strength may change over time.

Literature Review

Data Anchoring in Bitcoin

Bitcoin allows arbitrary data to be embedded within transaction structures.
Gao [3],proposes anchoring information by encoding data into fake P2PKH addresses.



The method permanently burns Bitcoin, making it economically inefficient.

Literature Review

Data Anchoring in Bitcoin

Bartoletti et al. [4] analyze data embedding techniques in Bitcoin and highlight their impact on the UTXO set and node performance.

Sward et al. [5] further evaluate these techniques in terms of performance and network health, concluding that OP_RETURN-based approaches are preferable because they do not bloat the UTXO set or burn Bitcoin.

However, this comes at the cost of limited data capacity per transaction.

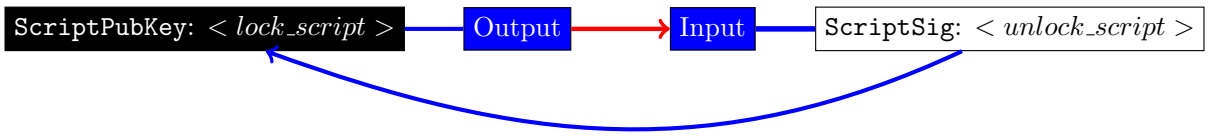
Literature Review

Data Anchoring in Bitcoin

Strehle and Steinmetz [6] provide an empirical analysis of OP_RETURN usage in Bitcoin transactions between 2018 and 2019.

Their study shows that OP_RETURN is used primarily by a small number of services (37), accounting for approximately 22% of the analyzed transactions and contributing about 10 GB of additional on-chain data.

This demonstrates that OP_RETURN-based anchoring is widely adopted but explicitly reveals embedded metadata at the protocol level.



Compiler sort in order: `<ScriptSig + ScriptPubKey>`
OP_RETURN embeds metadata directly in the transaction output script
`<op_push + string(80 bytes) + op_return>`

Literature Review

OP_RETURN

OP_RETURN is the most widely used mechanism for anchoring non-monetary data in Bitcoin. It is adopted by protocols such as Blockcerts [7] and OpenTimestamps [8].

However, OP_RETURN-based approaches exhibit several limitations:

- Embedded data is explicitly visible on-chain, limiting privacy.
- Data capacity per transaction is constrained by policy and fee costs.
- Merkle-based aggregation is not enforced at the protocol level and must be implemented off-chain.

Literature Review

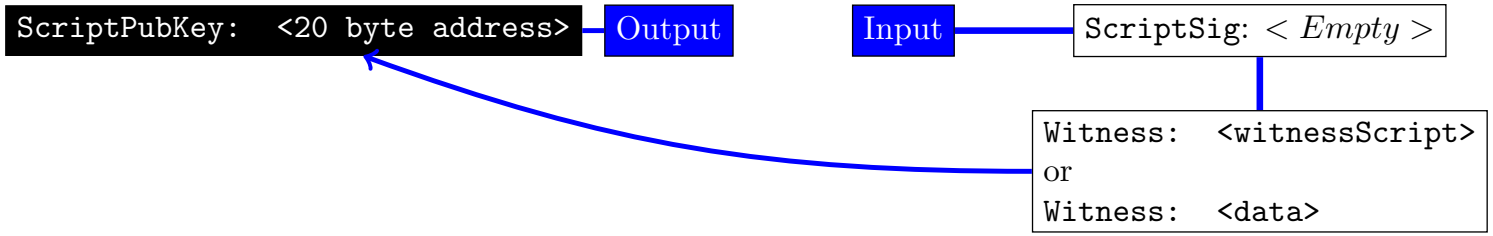
SegWit

Kedziora et al. [9] analyze the impact of the Segregated Witness (SegWit) soft fork after its deployment.

SegWit separates signature data from the transaction body into a dedicated witness structure, improving efficiency and security.

This design resolves transaction malleability and reduces effective transaction cost.

Additionally, anchoring data via the witness allows larger payloads without increasing UTXO set growth.



Compiler sort in order: `<Witness + ScriptPubKey>`

Literature Review

Solutions in Alternative Blockchains

Several works, including Wang [10], Li [11], Pennino [12], Palmisano [13], Mishra [14], and Harer [15], leverage smart contracts on Ethereum to attest information such as certificates, timestamps, and structured data.

These works show that:

- Smart contracts enable expressive and flexible certification logic.
- On-chain execution incurs high operational costs due to gas fees.
- Increased programmability introduces implementation complexity and security risks.

Literature Review

Example: Solutions for Academic Organizations

The work *Blockchain and Higher Education Diplomas* [16] notes that, as a radical innovation, blockchain adoption in academic contexts must overcome both technical barriers (scalability, error correction, system complexity) and cultural or social barriers (lack of standards and resistance to change).

A systematic literature review by Rustemi et al. [17] identifies that most existing academic certificate verification platforms are implemented on Ethereum and rely on smart contracts for service execution.

Remark

Verification can be performed autonomously, while certificate issuance and revocation remain challenging and often require third-party involvement.

Literature Review

New Cryptographic Tools

Bitcoin has evolved its protocol to improve security, privacy, and efficiency, with SegWit resolving transaction vulnerabilities [18].

Building on these foundations, Taproot integrates Schnorr signatures and Merkleized Abstract Syntax Trees (MAST), which have been extensively studied in the literature [19, 20, 21].

These cryptographic tools form the basis of the Taproot soft fork specified in BIP341, extending SegWit with more efficient and private spending conditions [22].

Taproot introduces a new transaction output type, Pay-to-Taproot (P2TR), activated on the Bitcoin network in November 2021.

This upgrade expands the design space for Bitcoin-native applications at the base layer.

Literature Review

Gap Identified

To the best of our knowledge, a complete Taproot-based protocol for decentralized document verification on Bitcoin has not been systematically studied.

Taproot provides:

- Improved privacy and efficiency through hidden script paths.
- Native support for compact commitments and batching via MAST.

Motivation

Problems with Bitcoin Before Taproot

Based on the literature, before Taproot building applications on Bitcoin involved several limitations:

Characteristic	**Fake Address**	**OP_RETURN**	**OpenTimestamps**	**Script Methods**	**Witness Methods**
Size (Bytes)	$N \times 20$	80	40	1000	4000
Privacy	No	Hash	Hash	Yes (until spend)	Yes (until spend)
Scalability	Limited	Merkle Tree	Merkle Tree	Complex	Complex
Construct Control	Yes	Yes	No	Complex	Complex
Impact UTXO set	High bloat	None	None	Low	Low
Cost (BTC)	Fee + Burn	Std	Aggregated	Low	Discount

Table: Comparison of anchoring methods on Bitcoin

Motivation

Trade-offs in Turing-Complete Networks

Other blockchain networks (e.g., Ethereum) adopt an account-based model to support expressive logic and smart contracts.

While this programmability enables sophisticated applications, it introduces several trade-offs:

- **Increased attack surface:** Greater implementation complexity expands the potential for vulnerabilities.
- **Consensus and execution overhead:** On-chain computation increases resource requirements and operational costs.
- **System fragility:** Complex execution environments are more susceptible to denial-of-service vectors and unintended behaviors.

Motivation

Taproot evolves the Bitcoin protocol by introducing features that enhance **efficiency, privacy, and security**.

Activated in 2021, it integrates **MAST** and **Schnorr signatures**, enabling more expressive spending conditions directly on the base layer.

These properties motivate revisiting document verification on Bitcoin within the design space enabled by Taproot.

Problem Statement

Research Questions

This work aims to answer the following questions:

- How can Taproot be used to build a decentralized document verification protocol on Bitcoin?
- How does using Merkle tree commitments in Taproot improve privacy and scalability during document verification?
- How can document revocation be supported within a Taproot-based verification protocol?

Problem Statement

Specific Objectives

To address the research questions, the following objectives are defined:

(a) To design a Merkle tree structure that enables the verification of document integrity.

(b) To design a mechanism for verifying issuer authenticity using digital signatures.

(c) To design a revocation tracking mechanism that allows the verification of document validity.

(d) To propose a proof of concept (PoC) of the proposed protocol on a Bitcoin test network.

(e) To estimate the protocol in terms of scalability, computational complexity, and on-chain costs measured in virtual bytes (vBytes).

Methodology

Overview

The proposed methodology is structured into two phases:

- **Phase I: Protocol and Architecture Design**
  Formal specification of a Taproot-based scheme for document batching and verification.
- **Phase II: Proof of Concept and Performance Estimations**
  Design of a Proof of Concept and analytical estimation of verification complexity and on-chain costs.

Methodology

Verification Goals

The proposed protocol supports three verification objectives:

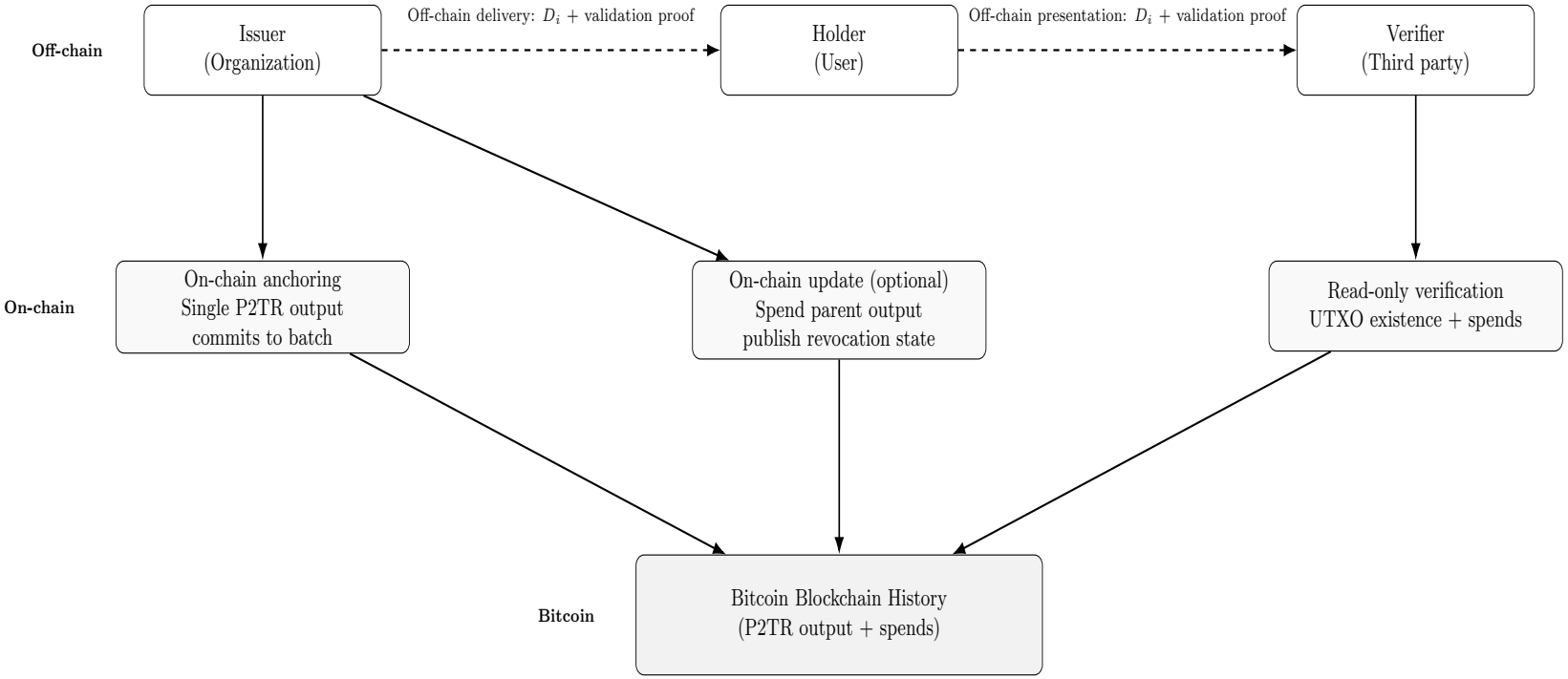
- **Integrity**: The document content has not been altered.
- **Authenticity**: The document was issued by the claimed organization.
- **Revocation**: The document has not been revoked.

Verification is performed **off-chain** using a **control file** provided together with the document.

The protocol is structured into two stages: **issuance** and **verification**.

Phase I

Issuance

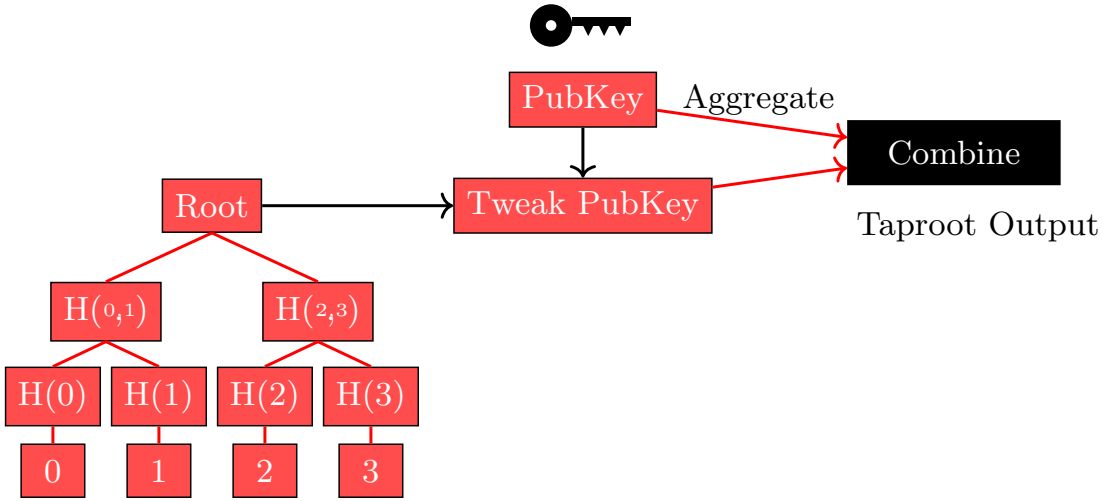


Phase I

Privacy via MAST

Privacy Preserved

Each script leaf contains the document hash and verification logic and is intentionally non-spendable. Only the script leaf related to the verified document is revealed. Other documents in the batch remain hidden.



Phase I

Cryptographic Binding (Taproot)

Script leaf structure

$$\mathsf{leaf}_i = \big(\, h_i \parallel \mathcal{L} \, \big), \quad h_i = \mathrm{SHA256}(D_i)$$

where h_i is the document hash and $\mathcal{L}$ represents non-spendable verification logic.

Merkle commitment

$$m = \mathrm{MerkleRoot}(\{\mathsf{leaf}_1, \ldots, \mathsf{leaf}_N\})$$

Cryptographic Binding (Taproot)

Taproot key derivation

$$Q = P_{\mathsf{btc}} + \mathrm{H}_{\mathsf{TapTweak}}(P_{\mathsf{btc}} \| m)\, G$$

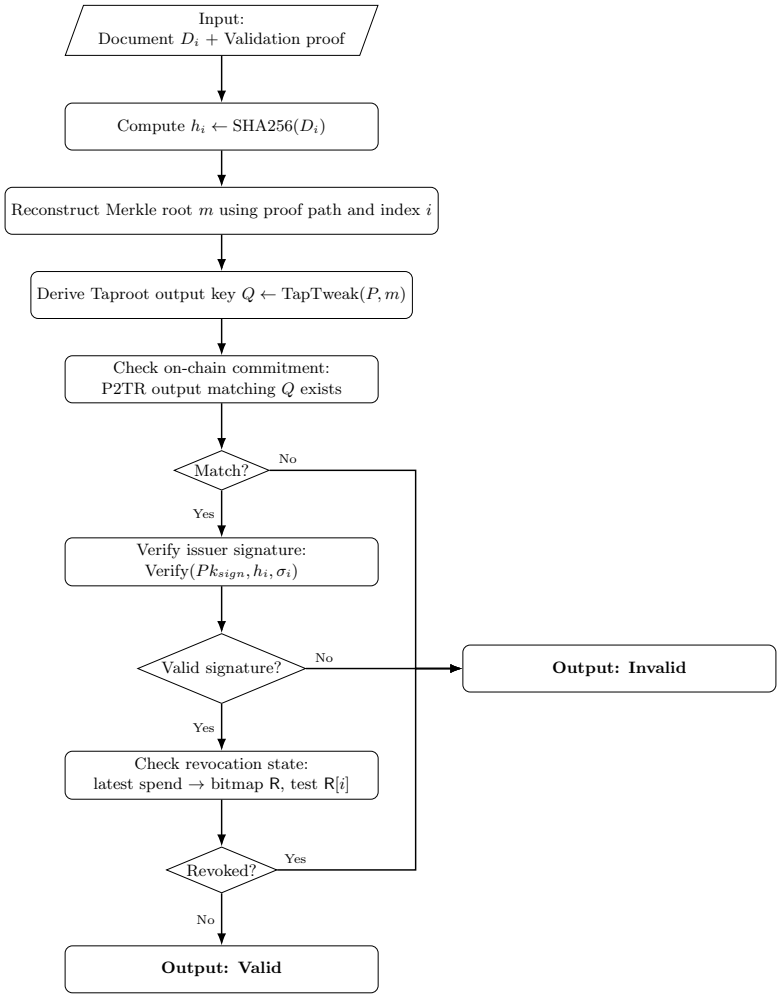
where P_{btc} is the issuer's Bitcoin public key used for Taproot output creation. **Document authenticity**

$$\sigma_i = \mathrm{Sign}(sk_{\mathsf{doc}},\, h_i), \quad \mathrm{Verify}(pk_{\mathsf{doc}},\, h_i,\, \sigma_i)$$

where $(pk_{\mathsf{doc}}, sk_{\mathsf{doc}})$ is the key pair used exclusively for document signing.

Phase I

Verification (Off-chain)



Phase I

Revocation (On-chain State Update)

Revocation is modeled as an **on-chain state update** rather than a modification of past data.

- The batch commitment is anchored in a P2TR output and cannot be edited after publication.
- To revoke, the issuer **spends** the current P2TR output and creates a new child output that commits to:
  - the same batch structure, and
  - a revocation state (e.g., a bitmap indexed by document position).
- Verifiers follow the spending chain to the **latest** output and read the revocation state to decide validity.
- Details on indexing and state encoding are **to be analyzed**.

Phase II

Proof of Concept and Evaluation Scope

This phase addresses objectives (d) and (e) through a controlled and analytical approach.

- **Proof of Concept (PoC) Design**:
  Specification of a reference workflow demonstrating issuance, verification, and revocation using the proposed protocol.
- **Experimental Context**:
  The PoC is designed for execution on Bitcoin **Signet/Testnet** environments to avoid real network costs.

Phase II

Metrics and Cost Analysis

The protocol is evaluated through analytical metrics focusing on scalability and on-chain cost.

Scalability

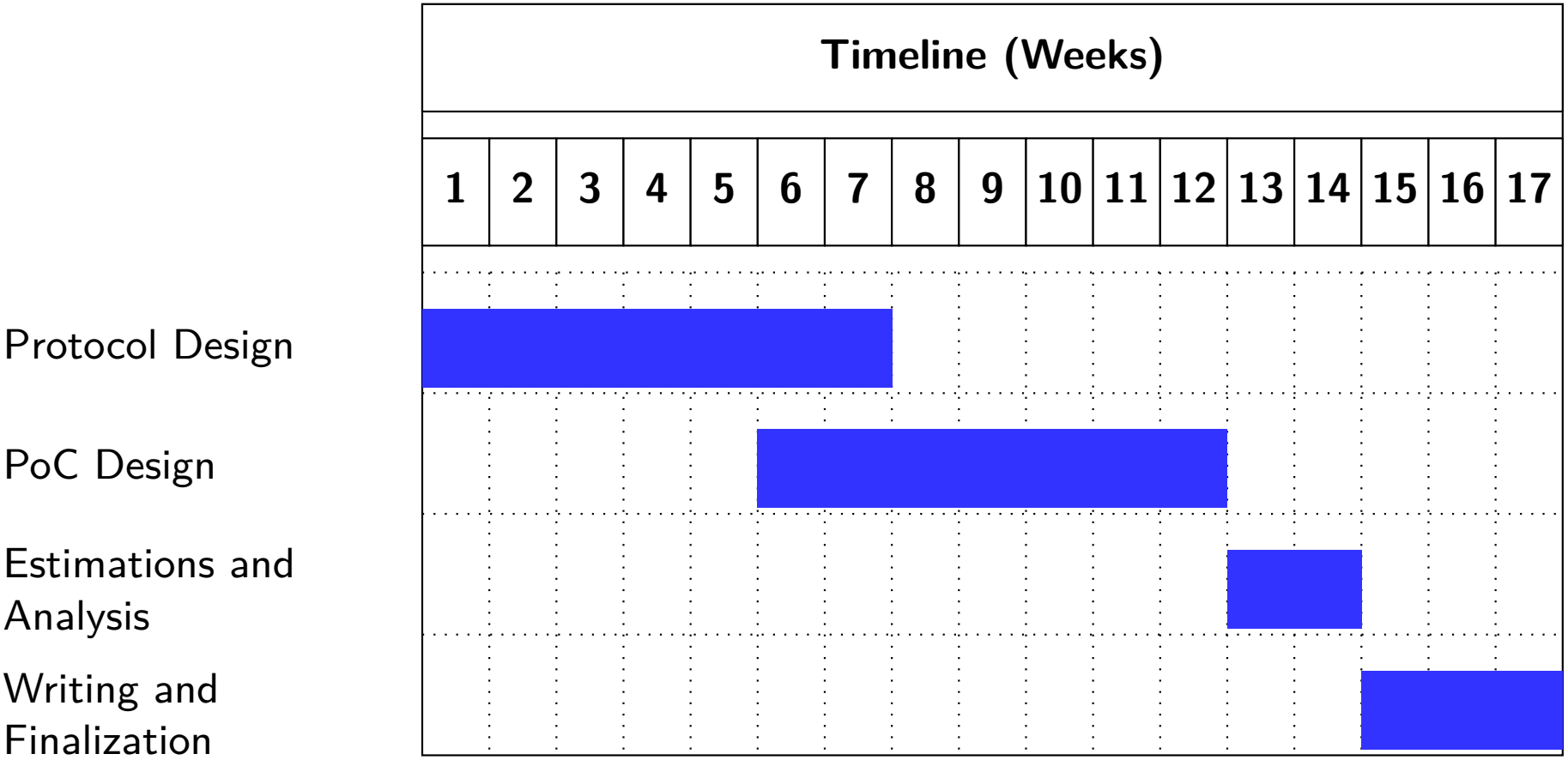
- Evaluated as a function of the batch size N.
- Verification complexity is bounded by $\mathcal{O}(\log N)$ due to Merkle tree inclusion proofs.

On-chain Cost

- Estimated in **vBytes**.
- Includes anchoring and revocation transactions.

The analysis highlights trade-offs between privacy, scalability, and network cost.

Timeline


Timeline (Weeks)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Protocol Design
PoC Design
Estimations and Analysis
Writing and Finalization

Conclusions

Expected Contributions

- This work motivates the use of decentralized certification protocols as an alternative to centralized document verification systems.
- The literature analysis found that Bitcoin-based approaches face limitations in scalability, privacy, and applicability to document verification.
- Existing solutions based on `OP_RETURN` and OpenTimestamps exhibit structural limitations at the protocol level.
- This work proposes a Taproot-based approach using Bitcoin-native cryptographic tools: Schnorr and Mast.

Conclusions

Expected Technical Capabilities

- The proposed protocol is designed to support integrity verification through hash commitments anchored in Merkle trees.
- Issuer authenticity is expected to be verified using crytographic digital signatures.
- Document revocation is addressed through transaction chaining and bitmaps on op return.
- Privacy is preserved by batching documents without revealing any information about unrelated documents.

Conclusions

Limitations and Future Work

- The proposed protocol does not address the semantic validity or correctness of document content.
- Document revocation remains under the control and responsibility of the issuing institution.
- Long-term document preservation and proof availability after issuance are not considered.
- The study of long-term preservation mechanisms remains an open area for future work.
- The design of incentives or penaltys mechanisms for misuse are left for future research.

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